01 / Project overview and scope

Customer Churn Prediction and Retention Dashboard

This college project estimates telecom churn with Logistic Regression, Random Forest and Decision Tree, and explores customer groups with a separate K-Means model. A saved-model Streamlit dashboard supports single and batch inference, comparisons and customer insights on a normal CPU.

Two learning tasks

Supervised classification learns a relationship between customer inputs and known churn labels. Unsupervised clustering groups feature vectors by distance without learning those labels. A cluster number is an arbitrary identifier, not a churn prediction or an ordered risk level. Silhouette and classification F1 cannot be ranked on one common scale.

Rubric clarification

Three supervised classifiers plus one unsupervised clustering model. Instructor approval may be needed if four supervised classifiers or four state-of-the-art models are required; full rubric compliance is not claimed.

Actual experiment

Run: 2026-10-16T10:08:29.285710+00:00. Raw customers: 3,150. Clean profiles: 2,797. Selected classifier: **Random Forest**, using validation F1 at a fixed 0.5 threshold. K-Means selected **k=2** from training features only.

Deliverables include the attributed dataset, reproducible splits, three fitted classifier pipelines, one clustering pipeline and PCA, saved metrics and profiles, interactive UI, tests, this ten-page submission report and a separate detailed Word learning guide. No paid API, database or GPU is needed.

The models estimate associations in one historical dataset. They do not establish that a proposed business intervention will retain a customer. Deployment remains prepared instructions, not a completed public service.

02 / Literature review of classification

[1] Prediction of Customer Churn Behavior in the Telecommunication Industry Using Machine Learning Models (2024)

Victor Chang, Karl Hall, Qianwen Ariel Xu, Folakemi Ololade Amao, Meghana Ashok Ganatra and Vladlena Benson. Algorithms 17(6), 231; author university repository and publisher record.

The telecom study compares machine learning classifiers including Decision Tree and Random Forest and discusses LIME/SHAP explanations. It motivates comparing a simple tree with an ensemble and inspecting several error metrics. Our project instead uses validation permutation importance and a frozen test protocol. Published performance on a different dataset is not an Iranian Churn benchmark; the paper does not establish that its data cleaning matches ours.

[2] A framework to improve churn prediction performance in retail banking (2024)

Joao B. G. Brito, Guilherme B. Bucco, Rodrigo Heldt, Joao L. Becker, Cleo S. Silveira, Fernando B. Luce and Michel J. Anzanello. Financial Innovation 10, 17; publisher version of record.

This banking framework emphasises RFM feature engineering and class-imbalance preprocessing rather than choosing an algorithm alone. It supports documenting preparation and minority-class evaluation. Unlike its transaction-scale sampling framework, our small aggregate-data project uses balanced classifier weights and no synthetic resampling. Its feature construction cannot be replicated from this CSV. Predictive ranking and segmentation remain different tasks in our implementation.

Comparison and implementation choice

The telecom comparison motivates keeping a tree baseline and an ensemble alongside Logistic Regression. The banking framework emphasises that feature construction and imbalance handling matter as much as the algorithm. Neither study establishes that its population or protocol matches this project. Their numerical scores are not treated as our results.

The implementation keeps a linear baseline, a regularised single tree and a modest forest. A small validation-only Decision Tree grid makes complexity choices explicit. The project does not describe these classical algorithms as four state-of-the-art classifiers. Research reviews here use verified accessible primary records and are not an exhaustive full-text systematic review.

03 / Literature review of segmentation

[3] Customer segmentation application based on K-Means (2024)

Jiaqi Zhao. Applied and Computational Engineering 47, 242-247; publisher record.

This paper reviews customer segmentation applications and introduces K-Means with RFM-style analysis. It motivates an interpretable grouping task alongside prediction, but does not justify treating a cluster ID as a churn label. Our telecom usage features are not transactional RFM. We evaluate distance-based groups using training silhouette, inertia and minimum size, and name them from training characteristics rather than assumed marketing value.

[4] Optimizing Customer Segmentation in Online Retail Transactions through the Implementation of the K-Means Clustering Algorithm (2024)

Desi Adrianti Awaliyah, Budi Prasetiyo, Rini Muzayanah and Apri Dwi Lestari. Scientific Journal of Informatics 11(2), 539-548; publisher record.

This retail study combines RFM-oriented customer analysis with K-Means and an elbow diagnostic, reporting three groups for its data. It supports examining inertia and feature profiles rather than reporting only a cluster count. Our reproducible training silhouette rule selects its own k and does not import the paper's three-group result. Feature scaling, category encoding and data populations differ, so numerical quality measures are not directly comparable.

[5] Analisis Pola Dan Prediksi Churn: Hybrid Segmentasi SOM+K-Means Dan Klasifikasi Machine Learning (2026)

Rizka Rahmadhani and Ermatita. JURIKOM 13(1), 335-346; publisher record, 28 February 2026.

The study separates a SOM plus K-Means grouping stage from supervised classification and uses cluster labels as classifier inputs. It highlights the distinction between segmentation and outcome prediction. Our architecture is deliberately simpler: no SOM, resampling or cluster-label feature is added to the churn classifiers. Segments are a parallel descriptive output. Churn rates are attached by split only after fitting and never choose k.

These studies support interpretable group profiling and separate grouping/prediction stages. Our clustering is parallel to churn classification; segment IDs are not classifier features. Published retail cluster counts or hybrid-model scores cannot be transferred to this telecom experiment. Sources were verified on 16 October 2026; complete DOI/links appear on page 10.

04 / Dataset and data cleaning

[6] Iranian Churn (2020), UCI Machine Learning Repository, DOI 10.24432/C5JW3Z, CC BY 4.0. Official source. Predictors aggregate the first nine months; Churn is observed at month twelve. Churn 1 means churn, 0 non-churn. No source timestamps independently confirm that timing.

Audit	Saved observation
Raw records / columns	3150 / 14
Missing cells / exact duplicates	0 / 300
Charge=10 quarantine / conflicting profile rows	7 / 83
Total removed / retained	353 / 2797
Raw non-churn / churn	2655 / 495

Retain ten documented inputs: failed calls, complaint flag, subscription months, charge band, call seconds, call count, SMS count, distinct called numbers, age group and tariff plan. Categories are complaint 0/1, charge 0-9, age 1-5 and tariff 1/2. Counts must be nonnegative integers; no upper limit is invented. Headers are normalised; valid extreme counts remain.

Seven official rows contain charge band 10 despite the notes specifying 0-9. The raw file is unchanged; row indices and quarantine reason are saved. It is unknown whether this is an error or an undocumented valid category. Remaining profiles with conflicting labels are removed, then identical retained-feature profiles are collapsed. This can merge distinct people and changes the population. Removal counts overlap the raw duplicate audit.

Status: 1 active / 2 non-active; conservatively excluded as an already-inactive account proxy, despite UCI observation-window statement.

Customer Value: Derived value with unpublished calculation; cannot independently audit its inputs or timing.

Age: Present in downloaded CSV but not defined separately in UCI variable notes; retain documented Age Group instead.

identifiers: Identifiers are never predictive features; absent in official downloaded CSV.

K-Means also excludes Churn and all identifiers. No missing values or synthetic records are introduced. Without IDs, profile isolation cannot prove person-level independence.

05 / Methods and leakage prevention

Partition	Profiles	Churn
Train	1957	295
Validation	420	63
Test	420	64

Seed-42 stratified splits preserve the existing approximately 70/15/15 row assignments. All classifiers and clustering share those partitions. Median/mode imputers, numeric scaling and categorical one-hot encoding fit training data only. No resampling is performed. Classifiers use balanced class weights; clustering uses no target or class weights.

Classifier configurations and selection

Logistic Regression: max_iter 1500. Random Forest: 180 trees, depth 12, minimum leaf 2, two CPU threads. Decision Tree: 12 combinations of max_depth [3,5,8], min_samples_split [2,10], min_samples_leaf [2,5]. Every candidate is fitted on training and ranked by validation churn F1, with ROC-AUC as tie-breaker. Selected tree settings: 8 depth, 2 split minimum, 2 leaf minimum.

The same validation criterion selects among the three final classifiers. Threshold 0.5 is fixed in advance. Selection is frozen before test evaluation. Models are not refitted on validation/test, and probabilities are not calibrated.

Separate K-Means geometry

Eight numeric/ordinal inputs median-imputed and standardised. Complains and Tariff Plan mode-imputed and full one-hot encoded without extra scaling; a category mismatch contributes squared distance 2. Ordinal band steps are treated equally. The geometry is a modelling assumption, not business value or risk.

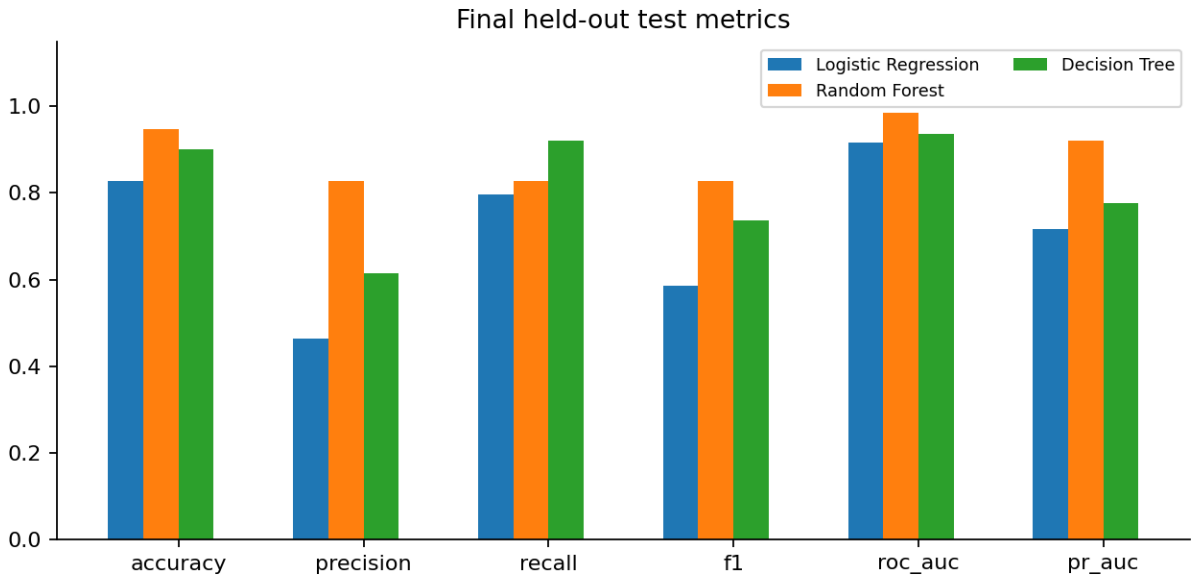
K-Means tries k=2-6 with n_init=10, max_iter=300 and seed 42. Training features only; discard candidates with a cluster below 2% of training rows when possible. Choose the smallest k within 0.01 of best eligible training silhouette. Inspect training inertia as an elbow diagnostic; it never uses churn.

Saved bundles include preprocessing, feature order, schema, seed, configuration, package versions and source hash; classifier bundles include threshold, and the cluster bundle includes training-fitted PCA. UI actions only load and predict.

06 / Actual classifier results

Classifier	Val F1	Test F1	Test AUC	Fit seconds
Logistic Regression	0.5646	0.5862	0.9162	0.018
Random Forest	0.8640	0.8281	0.9851	0.203
Decision Tree	0.7013	0.7375	0.9360	0.011

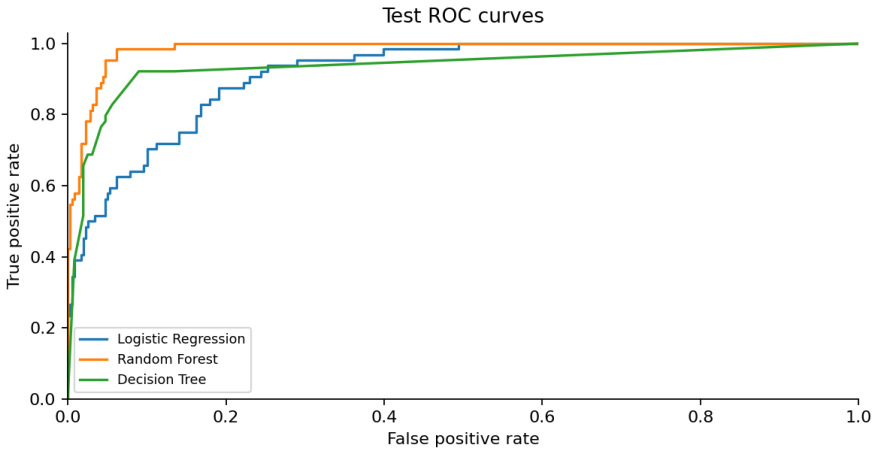
Classifier	Accuracy	Precision	Recall	Avg precision
Logistic Regression	0.8286	0.4636	0.7969	0.7178
Random Forest	0.9476	0.8281	0.8281	0.9200
Decision Tree	0.9000	0.6146	0.9219	0.7759

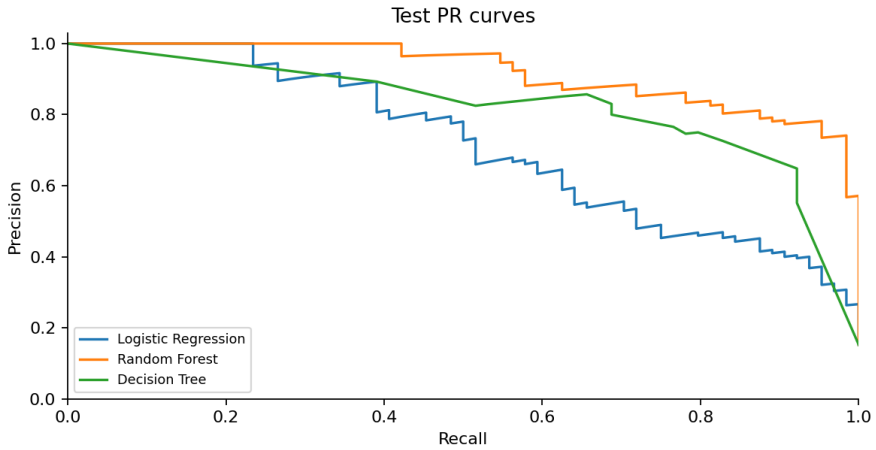


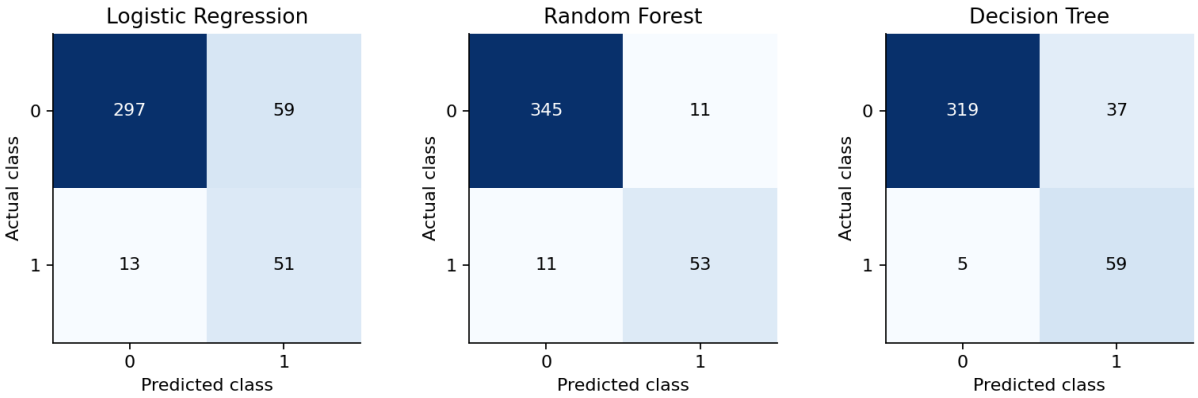
Random Forest remains selected from validation. These test figures are final reporting, not selection data. Fit time includes preprocessing and the selected estimator only; Decision Tree total candidate fitting took 0.105 seconds. Full candidate metrics and parameters are saved.

Accuracy is overall correctness. Precision is TP/(TP+FP), recall is TP/(TP+FN), and F1 is their harmonic mean. Average precision is a recall-weighted sum of precisions, not trapezoidal PR area. A majority-only classifier can have high accuracy while missing all churners. K-Means is intentionally absent from this classifier table.

07 / Classifier diagnostics and interpretation





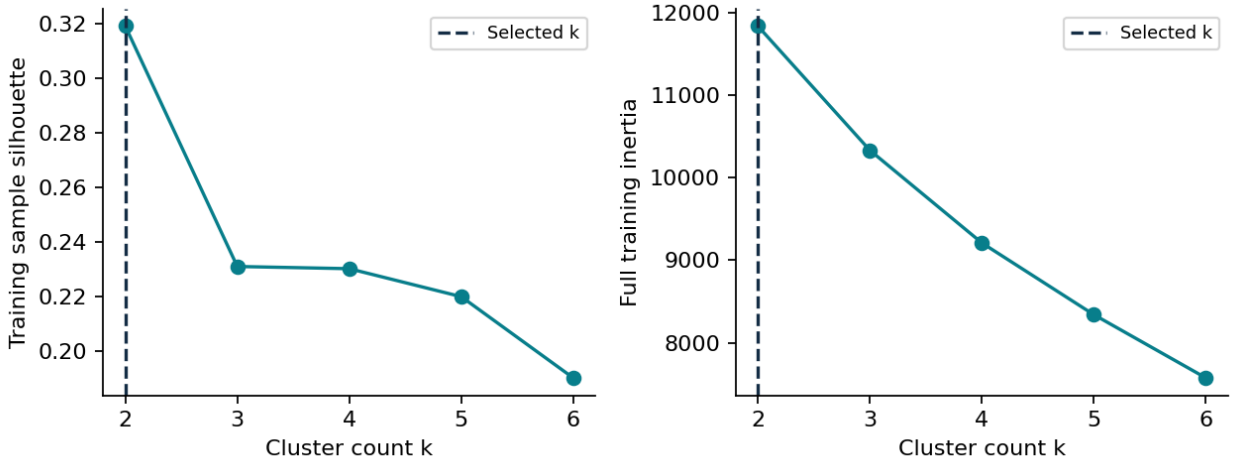


All curves and confusion matrices above use test data. Matrix rows are actual and columns predicted; 0 non-churn, 1 churn. Curves describe ranking tradeoffs without changing the deployed 0.5 threshold.

Validation permutation importance uses five F1-scored shuffles through each complete pipeline. The selected classifier's strongest mean decreases are Complains (0.209), Frequency of use (0.143), Seconds of Use (0.122). Correlation and sampling noise limit interpretation. Importance and form what-if changes do not establish causal retention effects.

08 / Actual K-Means evaluation

Selected k: **2**. Training silhouette: **0.3190** on a fixed 1,000-profile training-only sample. Full-training inertia: **11,834.9810**. Selected preprocessing + fit: 3.125 s; all candidate fit time: 3.669 s, excluding scoring/PCA analysis.



Split	Segment 0 size	Segment 1 size
Train	509	1448
Validation	104	316
Test	101	319

Segment 0: Higher call duration / Higher call frequency relative to overall training means.

Segment 1: Lower call duration / Lower call frequency relative to overall training means.

Names describe the two largest absolute standardised mean deviations among duration, SMS, call frequency, failures and tenure. IDs are arbitrary; these are neither risk ranks nor predictions. Validation/test/uploaded rows are assigned by saved nearest centroids. Split-specific feature means and transformed centroids are saved and shown in the UI.

Observed churn rates are attached by split only after fitting; they never choose k or names. PCA is fitted on training transformed features and used only for a labelled 2D projection. Inertia decreases with k, so the elbow is a visual diagnostic, not proof of a unique optimum. Euclidean geometry and encoded category weights are assumptions. Silhouette cannot be compared numerically with classifier F1.

09 / Application verification and deployment

Page	Final behaviour
Overview	Data KPIs, classifier/segment distinction and rubric clarification
Single Customer	Three classifier choices, grouped inputs, probability, threshold and what-if
Batch Prediction	Validated unlabelled upload, IDs, churn outputs and customer segments
Model Comparison	Three classifiers only; separate test/validation and metric/curve/confusion tabs
Customer Insights	Patterns, classifier importance and a separate segmentation tab

The professional navy/teal Streamlit design is preserved. Cached resources load trusted pipelines; no interaction trains a model. Batch validation preserves the existing exact schema, nonmissing finite values, documented categories and limits of 5 MB / 50,000 rows. Segment numbers and names are exported alongside probability/class without relabelling them as churn.

Tests cover schema, original split isolation, Decision Tree tuning, artifact loading, one/many predictions, training-only cluster transforms, exclusion of target/IDs/leakage fields, candidate selection, saved split assignments and all five pages. Actual execution outcomes and browser checks are recorded in reports/verification.md; this generator alone does not certify them. Final actual UI screenshots are used in the separate Word guide.

Run python -m src.train for intentional experiments, python -m pytest -q for checks, python -m src.report for this PDF and python -m streamlit run app.py for the UI in the pinned Python 3.13 environment. Follow README for Windows setup and local CSV fallback.

Community Cloud deployment is an instruction, not completed work: review/push an authorised GitHub repository, select app.py and Python 3.13, include trusted artifacts and locked dependencies, then verify logs, predictions, exports and mobile access. No public URL or physical-device test is claimed.

10 / Limitations conclusion and references

One historical dataset, conservative profile removal, unknown IDs/timestamps and the charge-band discrepancy limit validity. No calibration, uncertainty intervals or subgroup fairness study is claimed. Clusters depend on encoding and scale; names are descriptive, not causal or value rankings. New temporal data and controlled retention experiments are future work.

Random Forest won validation F1; K-Means selected 2 training-derived groups. Three supervised classifiers plus one unsupervised clustering model. Instructor approval may be needed if four supervised classifiers or four state-of-the-art models are required; full rubric compliance is not claimed.

[1] Victor Chang, Karl Hall, Qianwen Ariel Xu, Folakemi Ololade Amao, Meghana Ashok Ganatra and Vladlena Benson (2024). Prediction of Customer Churn Behavior in the Telecommunication Industry Using Machine Learning Models. DOI 10.3390/a17060231. Primary record.

[2] Joao B. G. Brito, Guilherme B. Bucco, Rodrigo Heldt, Joao L. Becker, Cleo S. Silveira, Fernando B. Luce and Michel J. Anzanello (2024). A framework to improve churn prediction performance in retail banking. DOI 10.1186/s40854-023-00558-3. Primary record.

[3] Jiaqi Zhao (2024). Customer segmentation application based on K-Means. DOI 10.54254/2755-2721/47/20241400. Primary record.

[4] Desi Adrianti Awaliyah, Budi Prasetiyo, Rini Muzayanah and Apri Dwi Lestari (2024). Optimizing Customer Segmentation in Online Retail Transactions through the Implementation of the K-Means Clustering Algorithm. DOI 10.15294/sji.v11i2.6137. Primary record.

[5] Rizka Rahmadhani and Ermatita (2026). Analisis Pola Dan Prediksi Churn: Hybrid Segmentasi SOM+K-Means Dan Klasifikasi Machine Learning. DOI 10.30865/jurikom.v13i1.9409. Primary record.

[6] Iranian Churn [Dataset] (2020), UCI. DOI 10.24432/C5JW3Z. CC BY 4.0. Official data.

[7] scikit-learn: K-Means documentation. [8] Streamlit: Community Cloud deployment. Primary research records checked 16 October 2026. Published scores on other datasets are not directly comparable with this experiment.